Table 1: TSP-50 Benchmark Results (10 random instances).

Method	**Mean Cost**	**Std**	**vs GT**
Nearest Neighbor	7.128	0.576	+20.8%
Christofides	6.428	0.374	+9.0%
C+2opt	5.900	0.312	*baseline*
DIFUSCO + 2-opt	5.918	0.313	+0.3%
DualOpt	**5.775**	0.300	**−2.1%**

GT = C+2opt with 5,000 iterations. DualOpt beats all methods including training labels.

Table 2: TSPLIB Standard Benchmark — Generalization Across Scales.

Instance	n	**NN**	**C+2opt**	**DIFUSCO**	**DualOpt**
eil51	51	20.6%	3.7%	5.4%	**1.0%**
berlin52	52	19.1%	4.6%	13.2%	**0.03%**
eil76	76	32.3%	7.3%	6.9%	**3.8%**
kroA100	100	26.2%	5.8%	10.0%	**3.8%**
ch150	150	25.5%	3.4%	**4.6%**	45.3%
tsp225	225	23.3%	3.1%	**3.5%**	36.8%
a280	280	22.1%	2.8%	7.7%	**6.7%**
pr1002	1002	21.8%	**3.9%**	7.1%	16.4%
Mean		23.9%	4.3%	6.9%	2.2%†

Gaps vs known OPT. Best per row in **bold**. †DualOpt mean for $n \leq 100$ only (degrades beyond due to fixed window sizes).
DIFUSCO trained **only** on TSP-50; generalizes to TSP-1002 (20× larger) with 7.1% gap.

Table 3: Ablation Study: Contribution of Inference Steps and 2-opt in DIFUSCO.

Configuration	**Mean Cost**	**vs GT**	**Time/inst**
10 steps, no 2-opt	6.724	+14.1%	0.17s
10 steps + 2-opt	5.896	+0.1%	0.17s
20 steps, no 2-opt	6.651	+12.9%	0.30s
20 steps + 2-opt	5.867	−0.4%	0.30s
50 steps, no 2-opt	6.697	+13.7%	0.72s
50 steps + 2-opt	5.866	−0.4%	0.72s

Key: 2-opt dominates quality. Raw diffusion worse than pure Christofides (9.0%). 10 steps + 2-opt ≈ 50 steps.

Table 4: DIFUSCO → DualOpt Hybrid Pipeline (TSP-50, 10 instances).

Method	Mean Cost	vs GT	vs Raw
DIFUSCO raw	6.696	+13.67%	*baseline*
DIFUSCO + 2-opt	5.969	+1.32%	−10.87%
DIFUSCO→DualOpt (ours)	**5.793**	**−1.67%**	**−13.50%**
C+2opt (baseline)	5.891	*baseline*	−12.03%

DualOpt reviser extracts 24% more improvement than 2-opt. Only method beating training labels.

Table 5: Pipeline Validation on TSPLIB ($n \leq 100$).

Instance	DualOpt	Ours	Δ
eil51	429.1 (0.73%)	437.4 (2.67%)	−1.9%
berlin52	**7,544 (0.03%)**	7,567 (0.34%)	−0.3%
eil76	562 (4.54%)	566 (5.14%)	−0.6%
kroA100	21,858 (2.71%)	**21,668 (1.81%)**	**+0.9%**

On kroA100, ours beats original DualOpt by 0.9 pp (cross-distribution robustness).

Table 6: Runtime & Quality Scaling ($n = 1000$).

Metric	C+2opt	DIFUSCO	DualOpt	LKH3
Time (s)	51.2	978.8	42.3	**1.2**
Gap vs OPT	3.9%	7.1%	16.4%	**0.7%**

LKH3: **43× faster** than C+2opt, **816× faster** than DIFUSCO at $n = 1000$, with best solutions.

Table 7: Five Improvements Attempted on DualOpt.

#	Improvement	Strategy	Outcome
1	Heatmap-Guided Reviser	Skip revision on confident windows	No effect (Δ = 0.00%)
2	**DIFUSCO→DualOpt**	**Replace initial tour with diffusion**	**SUCCESS (−1.67%)**
3	Adaptive Window Sizing	2-opt diagnostic → iteration budget	No improvement
4	Fragment Freezing	Lock edges where two solvers agree	Mixed: 4/10 improved, unstable
5	Destroy-and-Repair	Remove low-conf. edges, greedy reconnect	Degradation

Insight: All heatmap-guided attempts (#1,3,4,5) failed because TSP-50 is "solved" for DualOpt's reviser. Only #2 succeeded — it changes the *input* (initial tour), not the reviser's *internal process*.

Table 8: Algorithm Landscape: Six Methods Across Four Paradigms.

Method	Paradigm	Complexity	Key Property
Nearest Neighbor	Greedy heuristic	$O(n^2)$	20–30% gap (floor)
Christofides (1976)	Approximation	$O(n^3)$	1.5× guarantee
C+2opt	Improvement	$O(n^3)$	∼4.3% gap (baseline)
LKH3 (Helsgaun 2017)	Local search	$\sim O(n^{2.2})$	**0.4% gap (ceiling)**
DIFUSCO (NeurIPS'23)	Diffusion model	$O(n^2 d^2)$	Generative, 5.3M params
DualOpt (AAAI'25)	Divide + RL	$O(nk^2)$	Improvement, 710K params

Table 9: Scalability: Solution Quality & Runtime Across Instance Sizes (10 random instances each).

n	NN	Christofides	C+2opt	LKH3
Mean Tour Cost (lower = better)				
100	9.86	8.78	8.10	**7.82**
200	13.53	12.05	11.05	**10.72**
500	20.81	18.62	17.19	**16.63**
1000	28.72	26.11	23.96	**23.33**
Gap vs LKH3 (%)				
100	+26.2%	+12.4%	+3.6%	—
200	+26.2%	+12.4%	+3.2%	—
500	+25.1%	+11.9%	+3.3%	—
1000	+23.1%	+11.9%	+2.7%	—
Runtime per Instance				
100	1 ms	42 ms	52 ms	98 ms
200	5 ms	259 ms	322 ms	140 ms
500	38 ms	4.6 s	5.9 s	392 ms
1000	193 ms	37.3 s	51.2 s	**1.2 s**

LKH3 is 31× faster than Christofides and 43× faster than C+2opt at $n = 1000$, with best solutions. C+2opt maintains a steady 2.7–3.6% gap vs LKH3 across all scales.

Table 10: LKH3 as Gold Standard: Near-Optimal Across All TSPLIB Instances.

Instance	n	LKH3 Cost	Known OPT	Gap
eil51	51	430.0	426	0.94%
berlin52	52	7,544.4	7,542	0.03%
eil76	76	545.2	538	1.34%
kroA100	100	21,285.4	21,282	0.02%
ch150	150	6,530.9	6,528	0.04%
tsp225	225	3,859.0	3,916	−1.46%*
a280	280	2,587.8	2,579	0.34%
pr1002	1002	260,920.7	259,045	0.72%
Mean				**0.40%**

*tsp225: LKH3 found a better tour than the commonly cited optimum. Mean gap: 0.40%. LKH3 defines the speed-quality ceiling for all methods in this study.

Table 11: Improvement #4: Fragment Freezing — Per-Instance Analysis (TSP-50).

Inst.	Agreement	Original DualOpt	#4 Freezing	Δ	Outcome
1	56%	5.545	5.259	−5.16%	Best gain of any method
6	58%	5.346	5.277	−1.30%	Improved
7	58%	6.084	5.868	−3.55%	Improved
4	78%	5.426	5.426	0.00%	High agreement, no effect
9	56%	5.726	6.610	+15.43%	Catastrophic (wrong consensus)

Mean: +1.51% degradation. 4/10 instances improved. Agreement rate: 61.6% (avg).
When both solvers agree on *wrong* edges (∼40% of time), errors are locked in.
On TSPLIB: complete failure — DIFUSCO heatmap too poor for valid greedy merge (<0% OPT impossible).